

Critical Path Method (CPM)

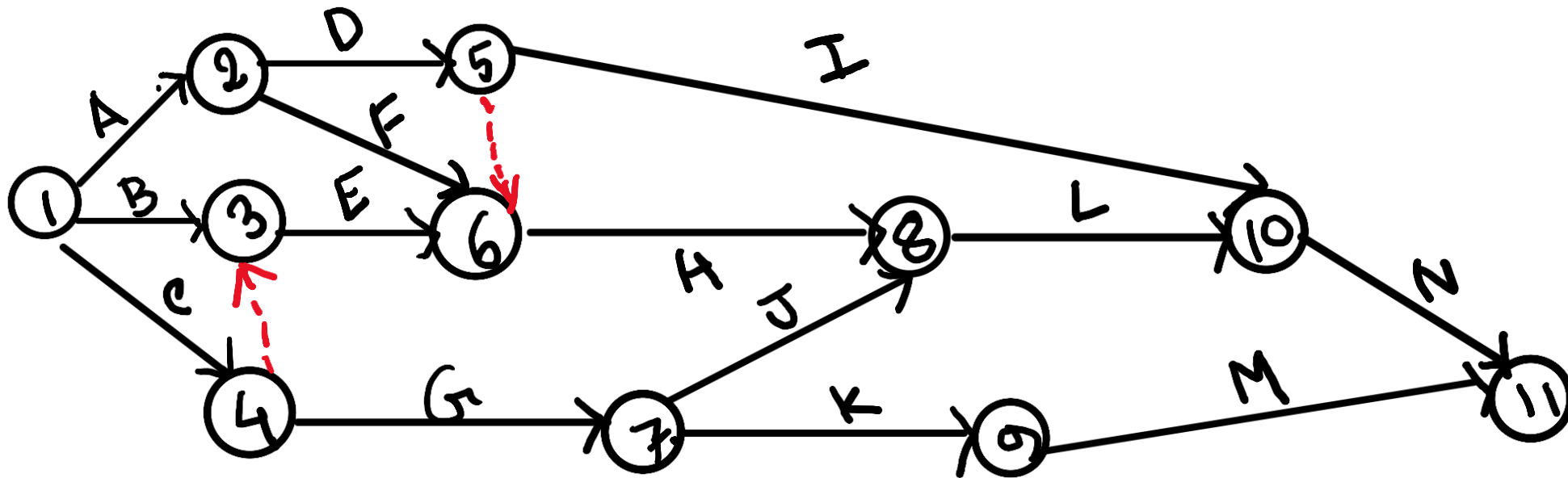
The Critical Path Method (CPM) is a powerful project scheduling and management technique developed in the late 1950s.

Basic Concepts and Terminology:

- **Activity:** A task or job consuming time and resources.
- **Event (Node):** A point marking the beginning or end of an activity.
- **Network:** A graphical representation of activities and events.
- **Duration:** The time required to complete an activity.

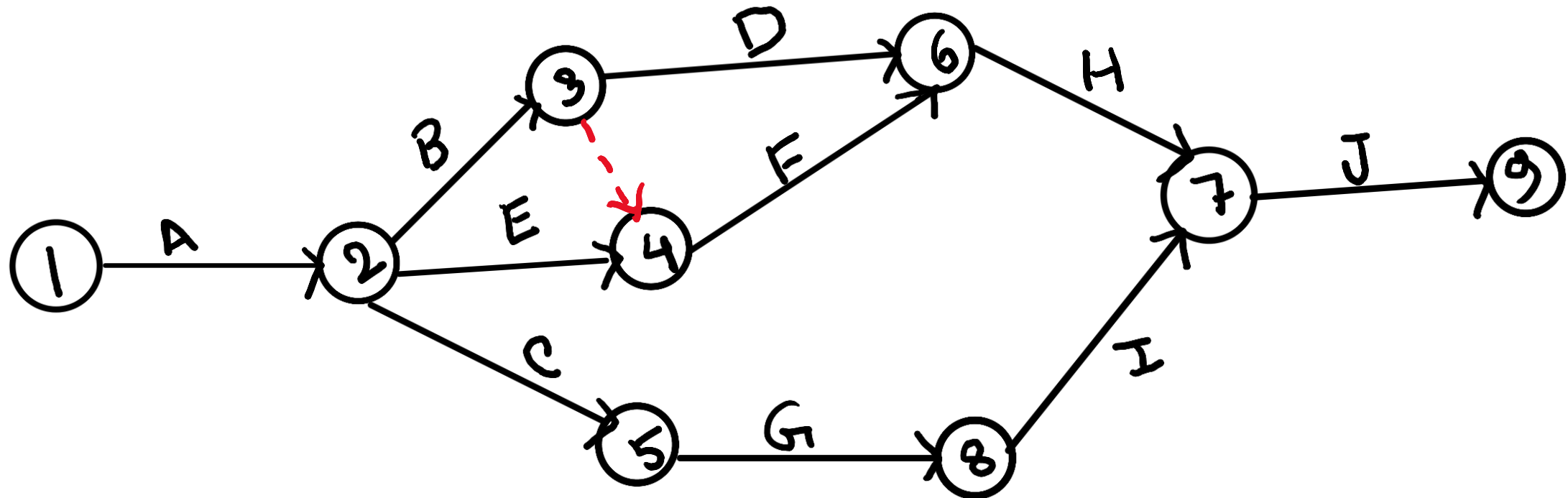
Q1: Draw a network diagram based on the following project:

Activity:	A	B	C	D	E	F	G	H	I	J	K	L	M	N
Preceding Activity:	—	—	—	A	B,C	A	C	D,E,F	D	G	G	H,J	K	I,L



Q2: Draw a network diagram based on the following project:

Activity:	A	B	C	D	E	F	G	H	I	J
Preceding Activity:	_	A	A	B	A	B,E	C	D,F	G	H,I



Earliest Occurrence Time (EOT) and Latest Occurrence Time (LOT)

The Earliest Occurrence Time is denoted by E_{S_i} for a event S_i .

The Latest Occurrence Time is denoted by L_{S_i} for a event S_i .

For any event (node) i , the earliest time at which it can occur is given by:

$$E_{S_i} = \max_{j \in P(i)} \{E_{S_{i-1}} + t_{ij}\}, E_{S_1} = 0.$$

Where t_{ij} = *time duration of the activity (i, j)*.

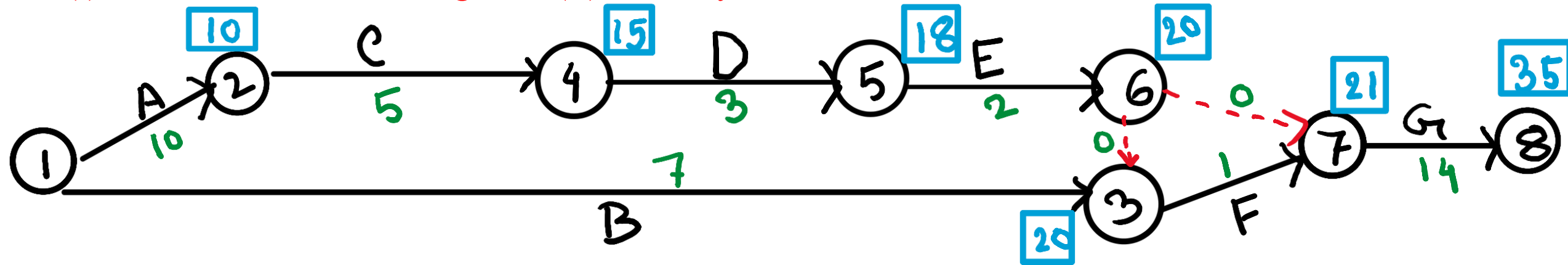
For any event (node) i , the latest time at which it can occur is given by:

$$L_{S_i} = \min_{j \in P(i)} \{L_{S_i} - t_{ij}\}, L_{S_l} = E_{S_1}.$$

Q2: Consider the following project:

Activity:	A	B	C	D	E	F	G
Preceding Activity:	—	—	A	C	D	B,E	E,F
Duration:	10	7	5	3	2	1	14

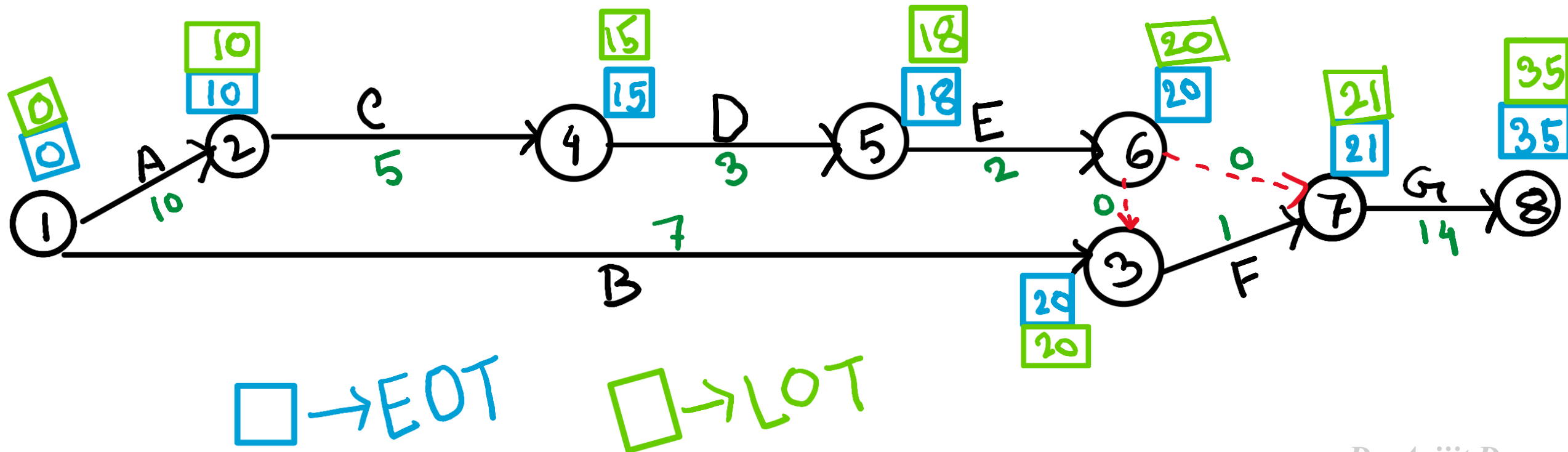
(i) Draw the network diagram, (ii) Identify the Earliest Occurrence Time at each event.



Q2: Consider the following project:

Activity:	A	B	C	D	E	F	G
Preceding Activity:	—	—	A	C	D	B,E	E,F
Duration:	10	7	5	3	2	1	14

(i) Draw the network diagram, (ii) Identify the Latest Occurrence Time at each event.



Critical Activity and Critical Path

Critical Activity: An activity (i, j) is called critical activity if it satisfies

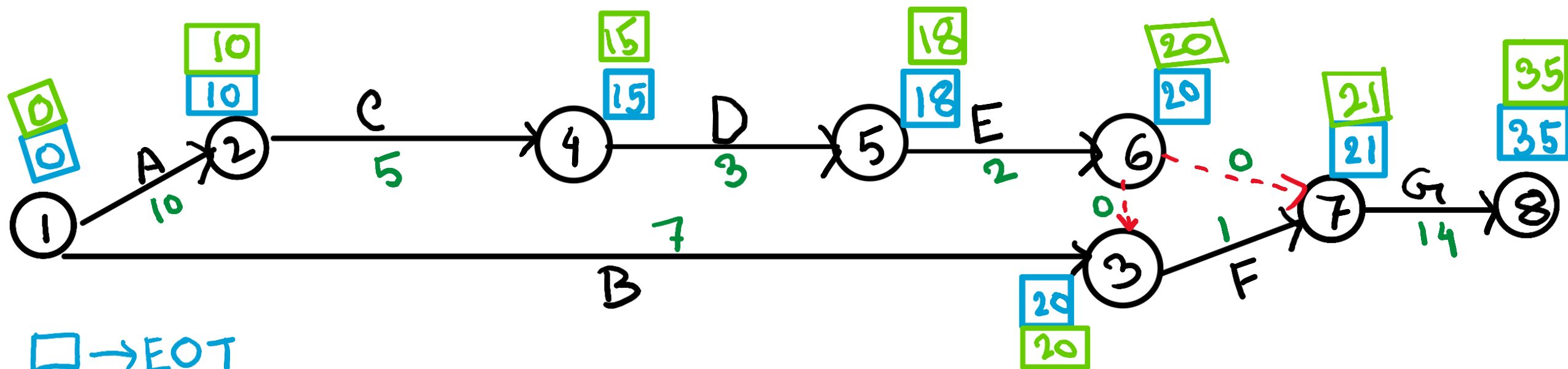
1. $E_{S_i} = L_{S_i}$
2. $E_{S_j} = L_{S_j}$,
3. $E_{S_j} - E_{S_i} = L_{S_j} - L_{S_i} = t_{ij}$.

Critical Path: A path, which comprises only the critical activities; spans entire network is called critical path.

Q2: Consider the following project:

Activity:	A	B	C	D	E	F	G
Preceding Activity:	—	—	A	C	D	B,E	E,F
Duration:	10	7	5	3	2	1	14

(i) Draw the network diagram, (ii) Identify the critical path.

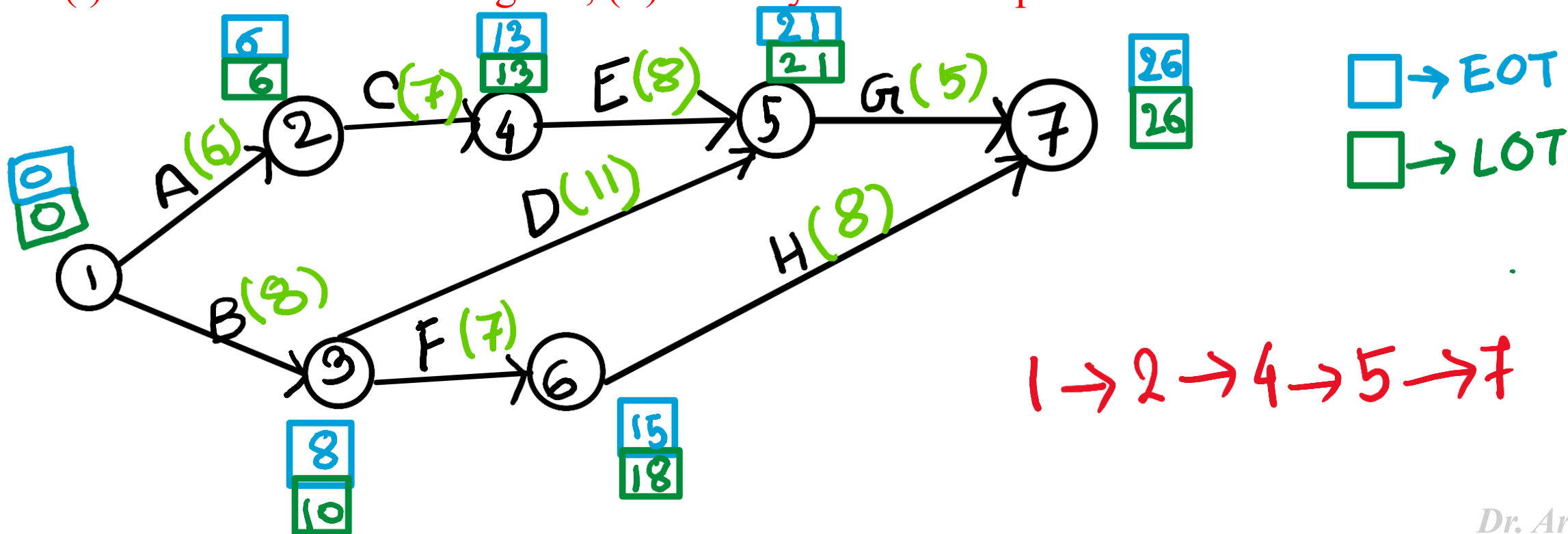


1 → 2 → 4 → 5 → 6 → 3 → 7 → 8

Q2: Consider the following project:

Activity:	A	B	C	D	E	F	G	H
Preceding Activity:	—	—	A	B	C	B	D,E	F
Duration:	6	8	7	11	8	7	5	8

(i) Draw the network diagram, (ii) Identify the critical path.

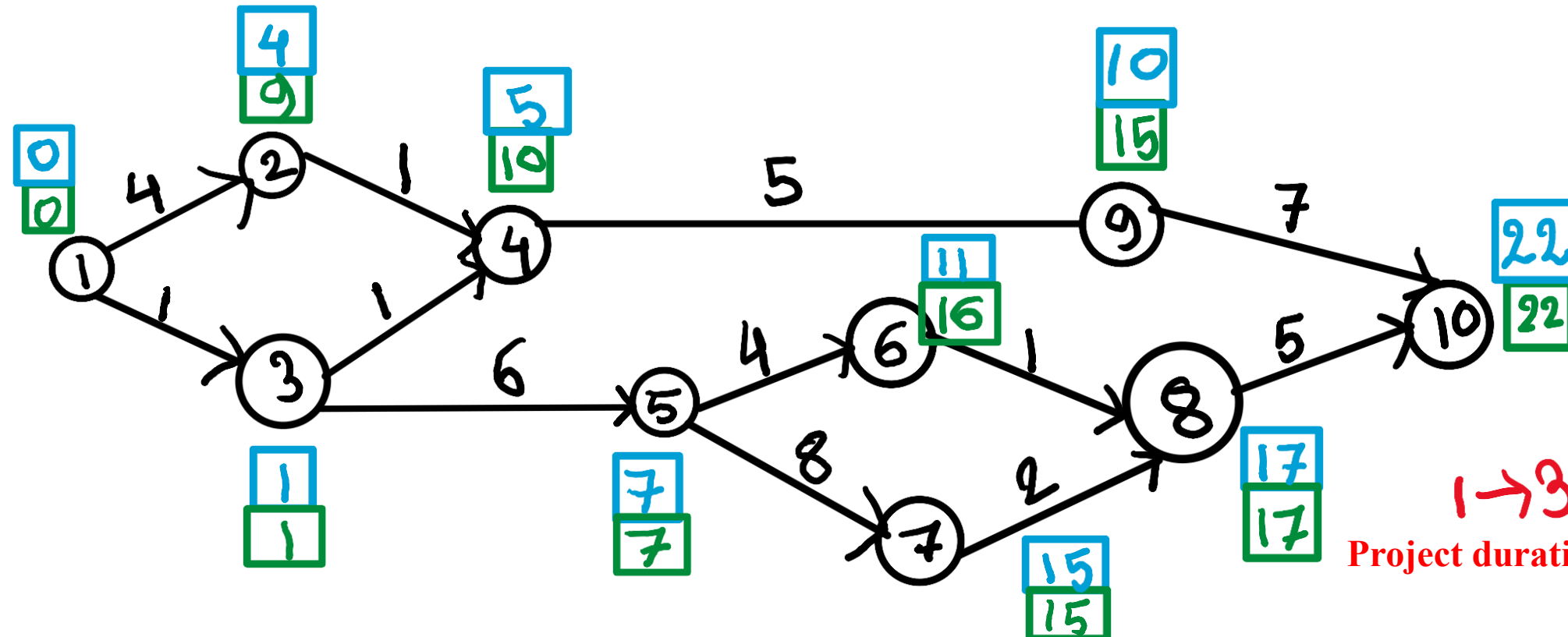


Q2: Consider the following project:

Activity: 1-2 1-3 2-4 3-4 3-5 4-9 5-6 5-7 6-8 7-8 8-10 9-10

Duration: 4 1 1 1 6 5 4 8 1 2 5 7

(i) Draw the network diagram, (ii) Identify the critical path, (iii) What is the project duration.



1 → 3 → 5 → 7 → 8 → 10

Project duration = 22 Dr. Arijit Das

Float (Slack) of an Activity and Event

Float: The Float is the length of time to which a non-critical activity and/or event can be delayed or extended without delaying the total project completion time.

Time Span: It is the time interval in which an activity is scheduled to finish. It is the interval given by (E_{S_i}, L_{S_j}) .

$$Time\ Span = L_{S_j} - E_{S_i}$$

Total Float: Total float is the maximum amount of time by which completion of an activity can be delay without affecting the overall duration of the project.

$$Total\ Float(TF_{ij}) = (L_{S_j} - E_{S_i}) - t_{ij} = Time\ Span - t_{ij}.$$

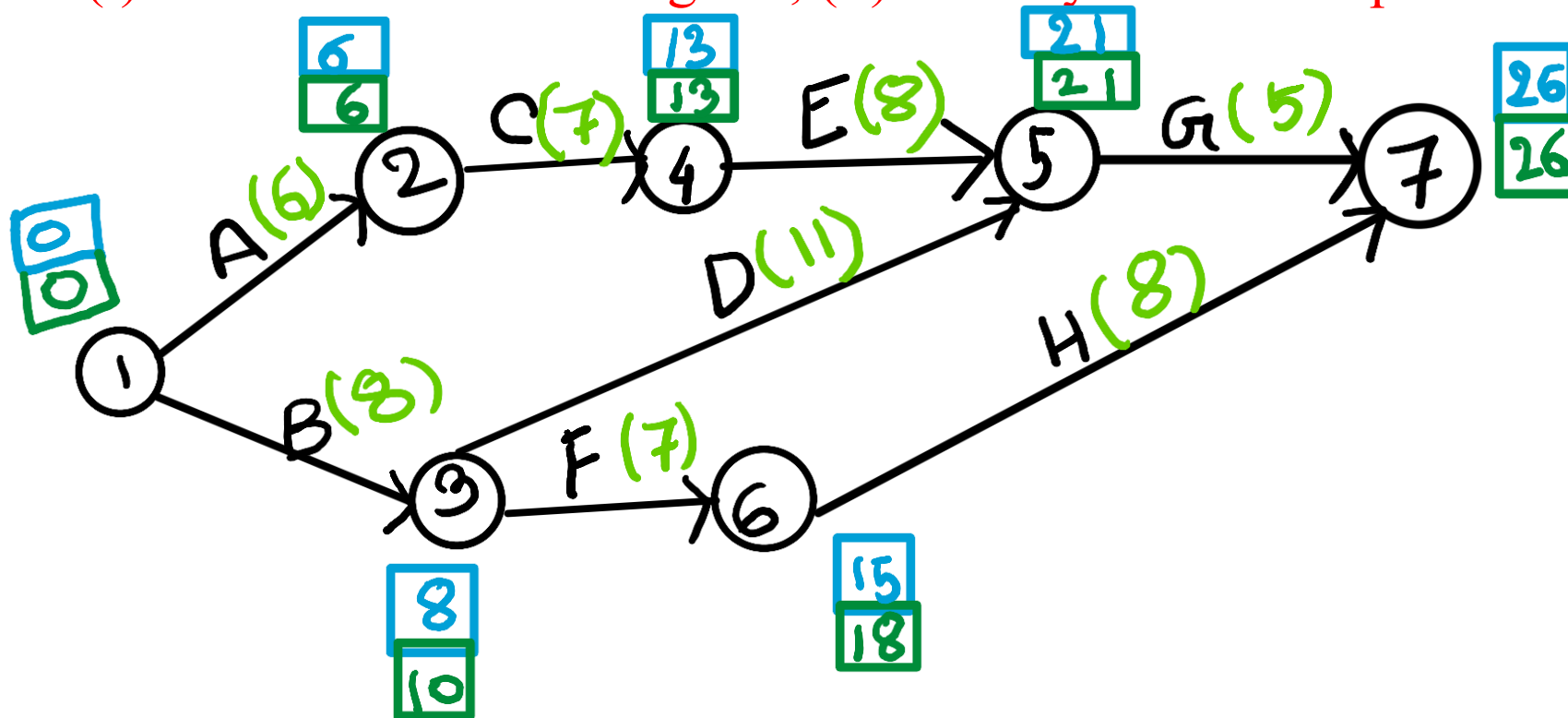
Free Float: Free float of an activity is the amount of time by which if an activity delayed it does not affect the earlies start of the succeeding activity.

$$Free\ Float(FF_{ij}) = (E_{S_j} - E_{S_i}) - t_{ij}.$$

Q2: Consider the following project:

Activity:	A	B	C	D	E	F	G	H
Preceding Activity:	—	—	A	B	C	B	D,E	F
Duration:	6	8	7	11	8	7	5	8

(i) Draw the network diagram, (ii) Identify the critical path.



Activity	Time interval	Time Span	TF	FF
A (1,2)	(0, 6)	6	0	0
B (1,3)	(0, 10)	10	2	0
C (2,4)	(6, 13)	7	0	0
D (3,5)	(8, 21)	13	2	2
E (4,5)	(13, 21)	8	0	0
F (3,6)	(8, 18)	10	3	0
G (5,7)	(21, 26)	5	0	0
H (6,7)	(15, 26)	11	3	3

Project Crashing

Project Crashing is a technique in project management used to reduce the total project duration **by shortening the time of one or more critical activities**. This reduction is achieved **by allocating additional resources such as labor, equipment, or cost**.

To check the per unit increase in cost when an activity is crashed, we need to calculate the slope of the activity, which is given by

$$\textbf{Slope} = \frac{(C_c - C_N)}{T_N - T_C}.$$

To find all possible Crash Limits (i.e. , Limits of all activities that can be reduced

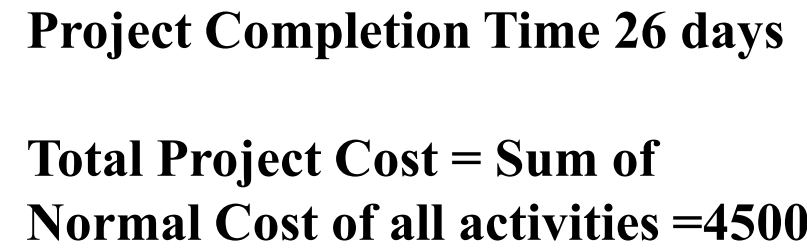
$$\textbf{Crash Limit} = T_N - T_C.$$

Note that, If an activity has a Crash Limit of 0. Then it means it cannot be reduced further, since it would lead to Slope being infinity due to division by 0.

Q2: Consider the following project:

Project				
Activity	Normal Duration	Normal Cost	Crash Duration	Crash Cost
(1-2)	6	300	5	400
(1-3)	8	400	6	600
(2-4)	7	400	5	600
(4-5)	8	1000	4	1350
(3-5)	11	500	5	800
(3-6)	7	400	6	500
(5-7)	5	1000	3	1400
(6-7)	8	500	5	950

- Draw the network diagram and find the project completion time and cost.
- Find most economical schedule if the project is to be complete in 23 days.
- Find the optimal (minimum) project duration and associated cost.



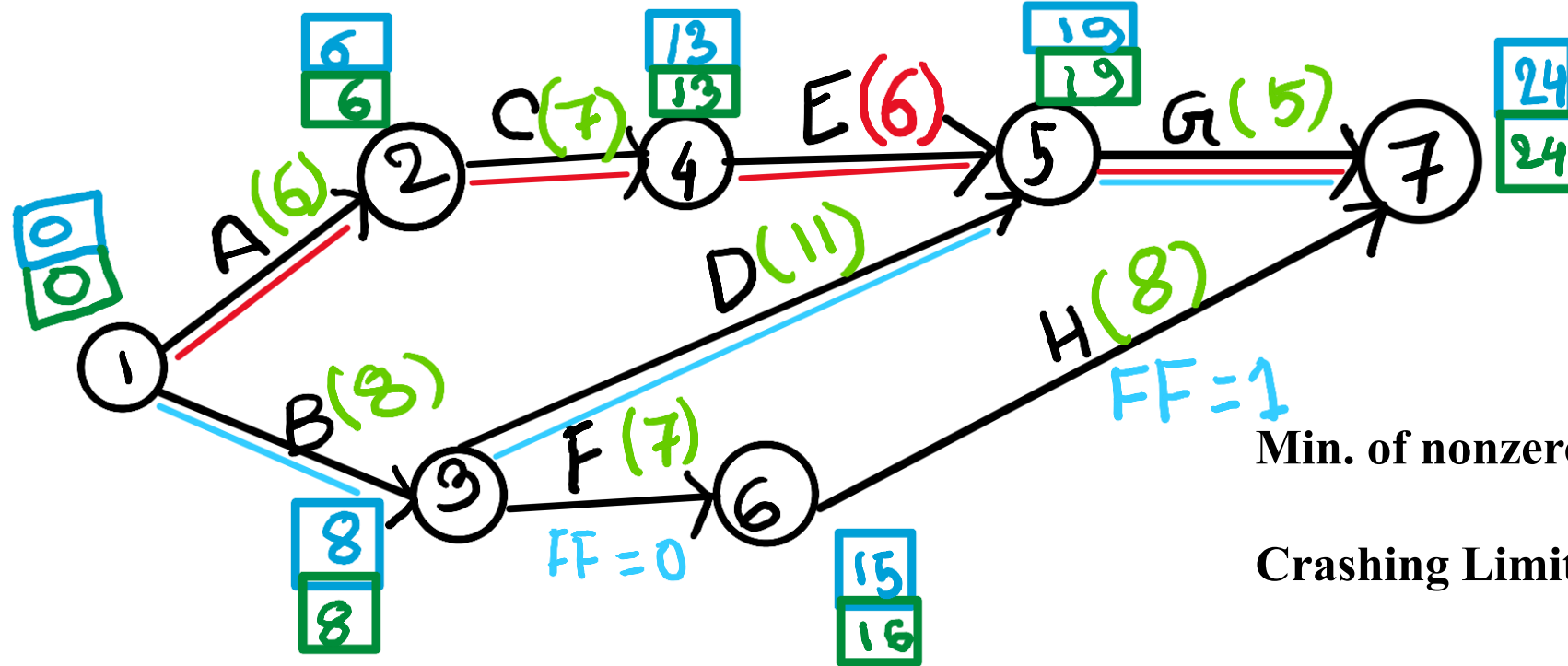
Min. of nonzero FF (FF Limit) = $\min\{2, 3\} = 2$

Crashing Limit (for activity with min slope) = 4

**Compression Limit = $\{FF\ Limit, Crash\ limit\} =$
 $= \min\{2, 4\} = 2$**

Therefore, we can reduce the duration of activity E(4, 5) by 2 days.

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Project Completion Time 24 days

Project Cost = 4500 + 87.5 (2) = 4675

Min. of nonzero FF (FF Limit) = $\min\{0, 1\} = 1$

Crashing Limit (for activity with min slope) = 2, 6

Compression Limit = $\{FF \text{ Limit}, Crash \text{ limit}\} = \min\{1, 2, 6\} = 1$

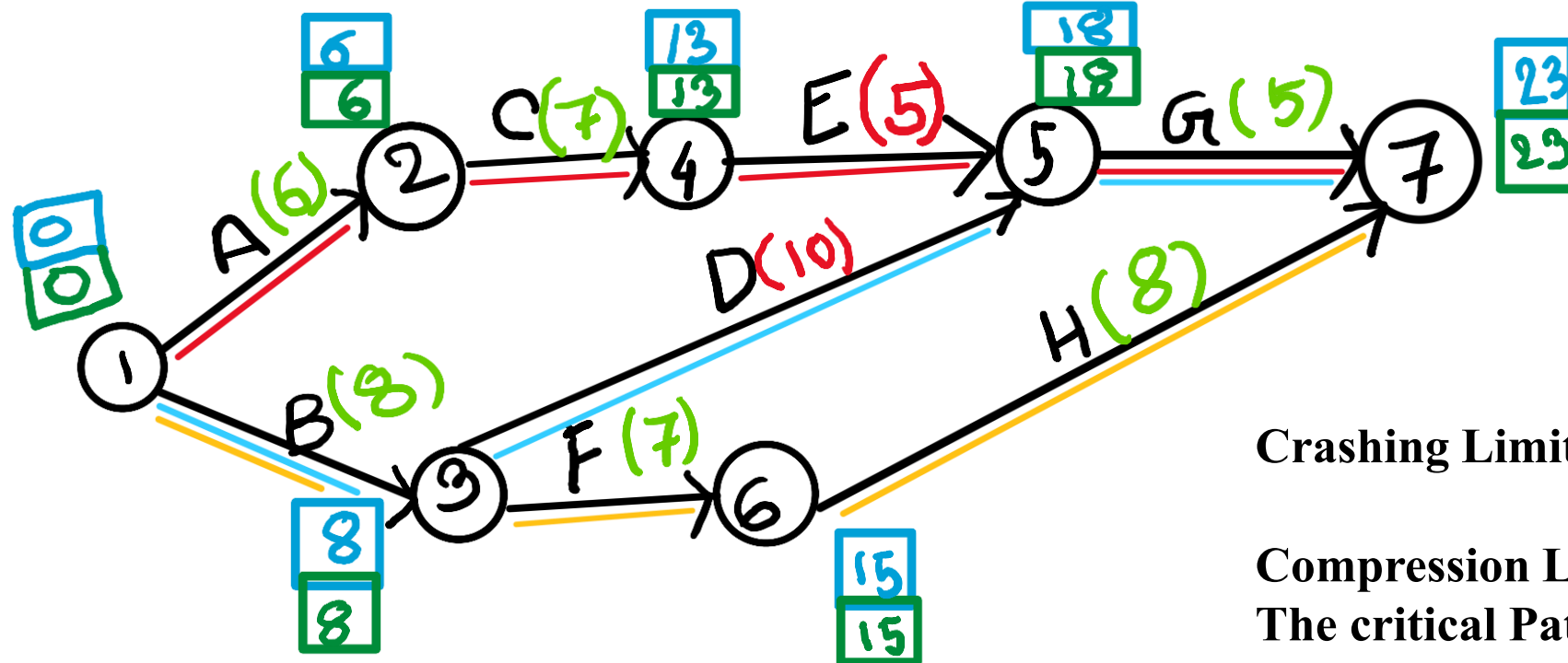
Therefore, we can reduce the duration of activity E(4, 5) and D(3, 5) by 1 days.

Note that, the common activity G (5, 7) can be selected for reduction if

Slop of the common act. < sum of the slop of individual activity. i.e., if

Slope of (5, 7) < slope of (4, 5) + slope of (3, 5), which is not correct so can not select activity G (5, 7). Dr. Arijit Das

Activity	T_N	C_N	T_C	C_C	Slope	Crash Limit
(1-2)	6	300	5	400	100	1
(1-3)	8	400	6	600	100	2
(2-4)	7	400	5	600	100	2
(4-5)	8 6	1000	4	1350	87.5	2
(3-5)	11	500	5	800	50	6
(3-6)	7	400	6	500	100	1
(5-7)	5	1000	3	1400	200	2
(6-7)	8	500	5	950	150	3



Project Completion Time 23 days

$$\text{Project Cost} = 4675 + 87.5(1) + 50(1) = 4812.5$$

Crashing Limit (for act. with min slope) = 1, 5, 1

Compression Limit = $\min\{1, 5, 1\} = 1$

The critical Paths are:

$1 \rightarrow 2 \rightarrow 4 \rightarrow 5 \rightarrow 7$

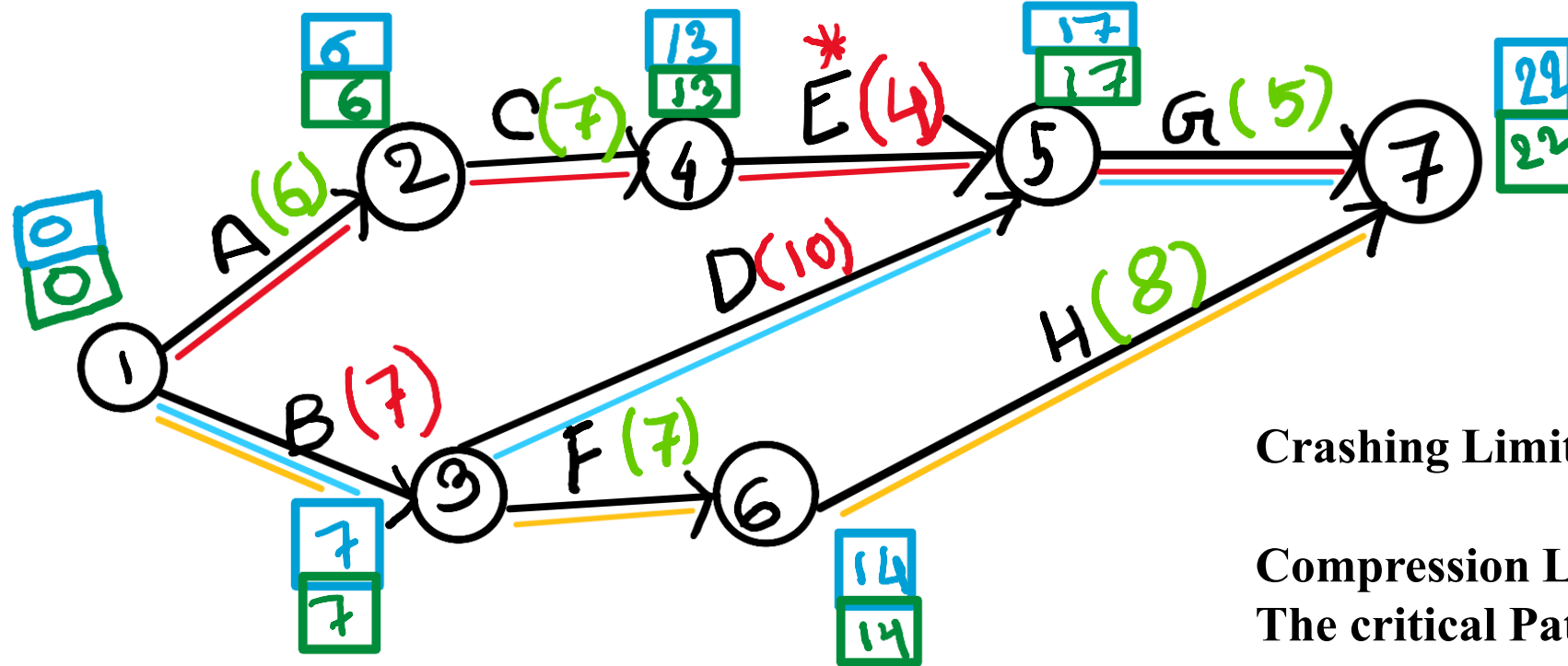
$1 \rightarrow 3 \rightarrow 5 \rightarrow 7$

$1 \rightarrow 3 \rightarrow 6 \rightarrow 7$

Slope of (1, 3) < slope of (3, 5) + slope of (3, 6)

Therefore, we can reduce the duration of activity B(1, 3) and E(4, 5) by 1 days.

Activity	T_N	C_N	T_C	C_C	Slope	Crash Limit
(1-2)	6	300	5	400	100	1
(1-3)	8	400	6	600	100	2
(2-4)	7	400	5	600	100	2
(4-5)	8 6-5	1000	4	1350	87.5	1
(3-5)	11 10	500	5	800	50	5
(3-6)	7	400	6	500	100	1
(5-7)	5	1000	3	1400	200	2
(6-7)	8	500	5	950	150	3



Crashing Limit (for act. with min slope) = 1, 5, 1

Compression Limit = {1, 5, 1} = 1.

The critical Paths are:

1 → 2 → 4 → 5 → 7

1 → 3 → 5 → 7

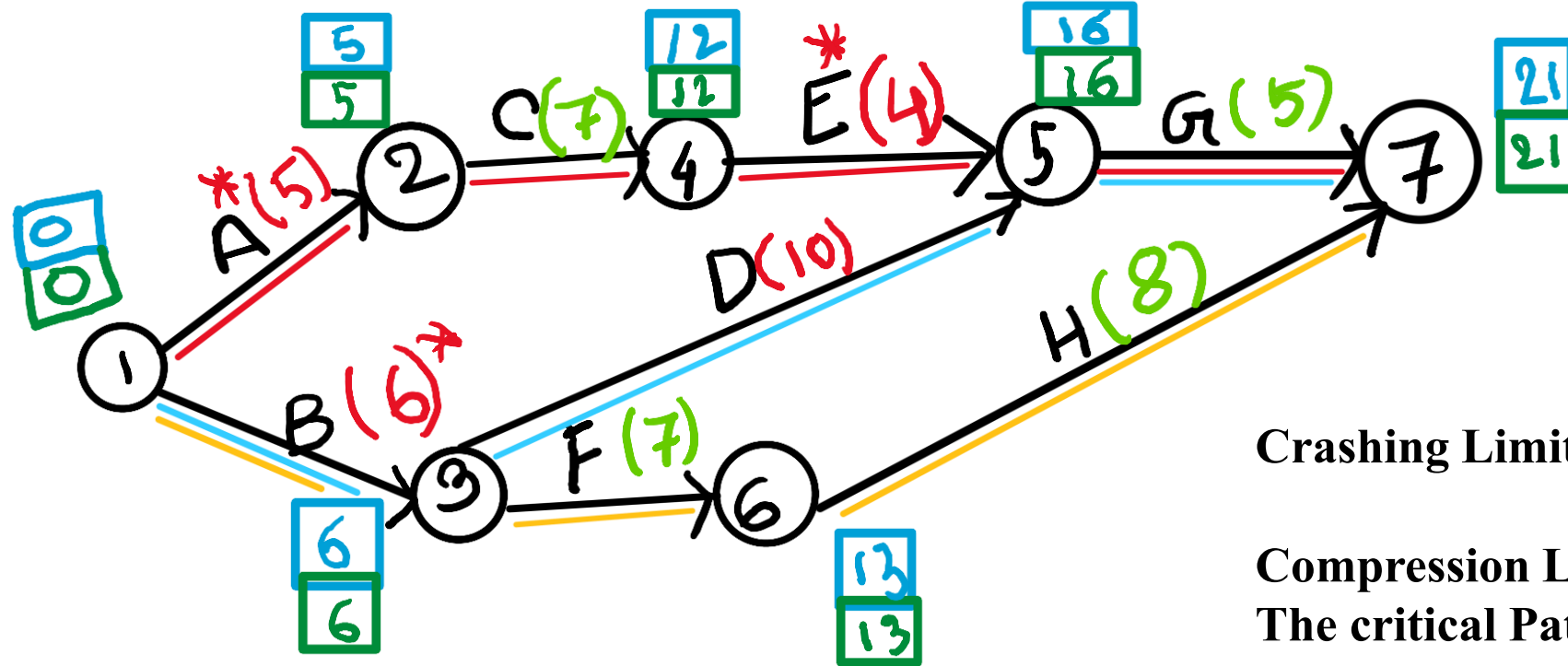
1 → 3 → 6 → 7

Slope of (1, 3) < slope of (3, 5) + slope of (3, 6)

From the critical path 1 → 2 → 4 → 5 → 7 we can any of the activity from (1, 2) and (2, 4). We choose (1, 2) for reduction.

Therefore, we can reduce the duration of activity B(1, 3) and A(1, 2) by 1 days.

Activity	T_N	C_N	T_C	C_C	Slope	Crash Limit
(1-2)	6	300	5	400	100	1
(1-3)	8 7	400	6	600	100	1
(2-4)	7	400	5	600	100	2
(4-5)	5 4	1000	4	1350	87.5	0 *
(3-5)	11 10	500	5	800	50	5
(3-6)	7	400	6	500	100	1
(5-7)	5	1000	3	1400	200	2
(6-7)	8	500	5	950	150	3



Project Completion Time 21 days

Project Cost = $5000 + 100 + 100$
 $= 5200$

Crashing Limit (for act. with min slope) = 2, 5, 1

Compression Limit = $\{2, 5, 1\} = 1$.

The critical Paths are:

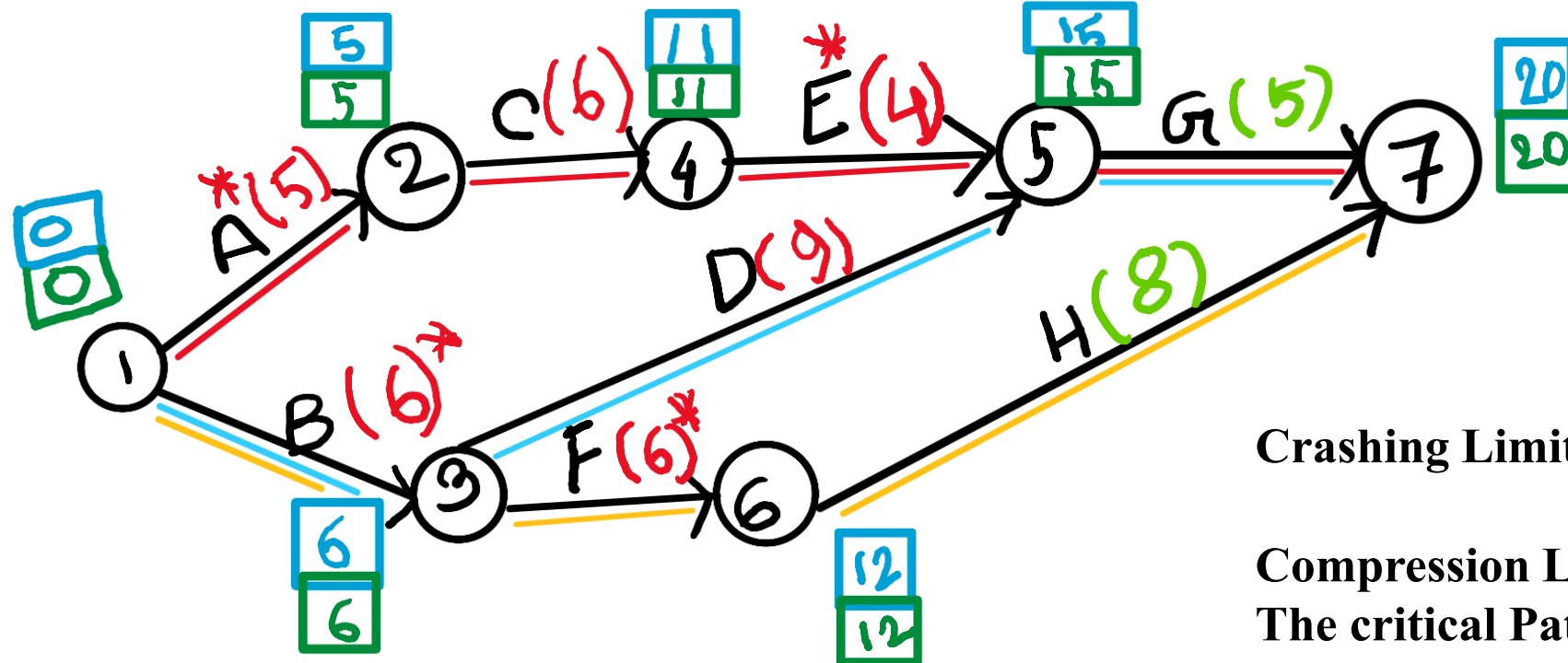
$1 \rightarrow 2 \rightarrow 4 \rightarrow 5 \rightarrow 7$

$1 \rightarrow 3 \rightarrow 5 \rightarrow 7$

$1 \rightarrow 3 \rightarrow 6 \rightarrow 7$

Therefore, we can reduce the duration of activity C(2, 4), D(3, 5) and F(3, 6) by 1 days.

Activity	T_N	C_N	T_C	C_C	Slope	Crash Limit
(1-2)	6 5	300	5	400	100	0 *
(1-3)	7 6	400	6	600	100	0 *
(2-4)	7	400	5	600	100	2
(4-5)	5 4	1000	4	1350	87.5	0 *
(3-5)	11 10	500	5	800	50	5
(3-6)	7	400	6	500	100	1
(5-7)	5	1000	3	1400	200	2
(6-7)	8	500	5	950	150	3



Project Completion Time 20 days

Project Cost = 5200 + 100 + 50 + 100
= 5450

Crashing Limit (for act. with min slope) = 1, 4, 3

Compression Limit = {1, 4, 3} = 1.

The critical Paths are:

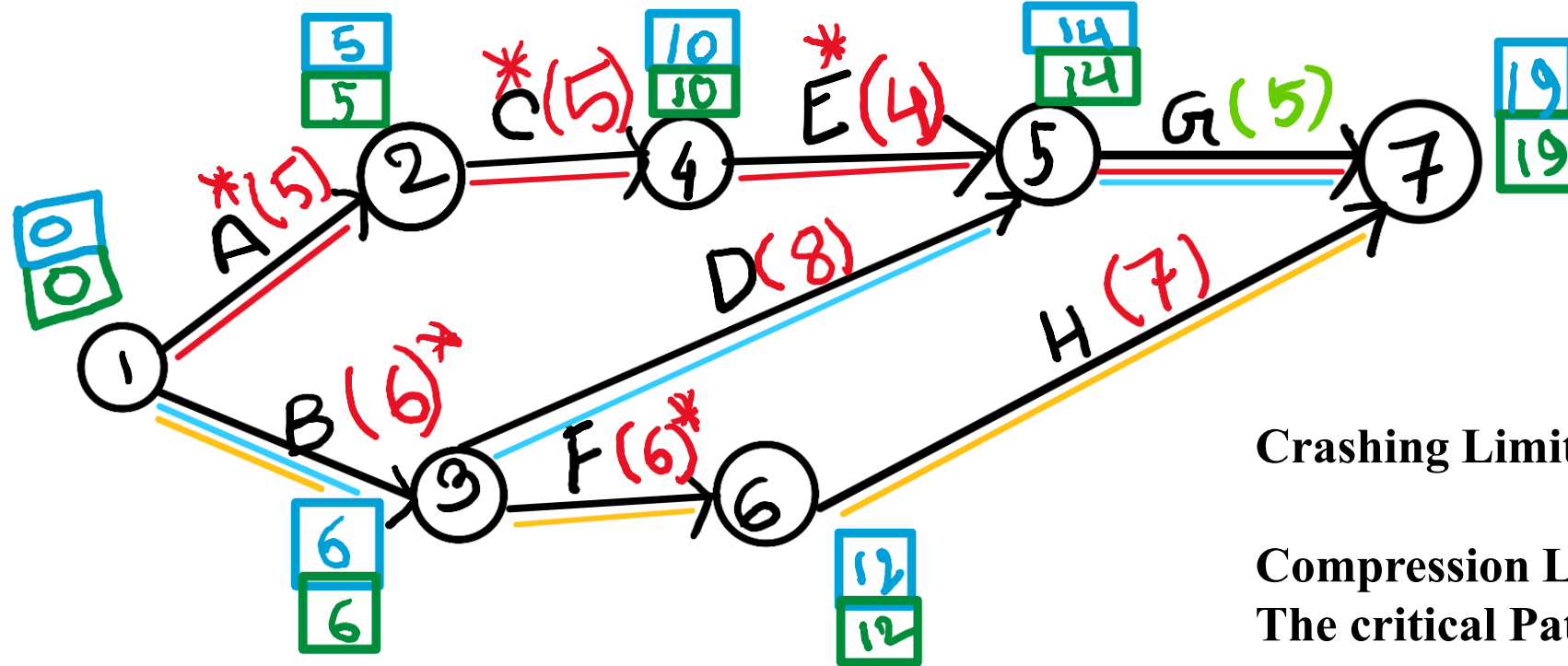
1 → 2 → 4 → 5 → 7

1 → 3 → 5 → 7

1 → 3 → 6 → 7

Therefore, we can reduce the duration of activity C(2, 4), D(3, 5) and H(6, 7) by 1 days.

Activity	T_N	C_N	T_C	C_C	Slope	Crash Limit
(1-2)	6 5	300	5	400	100	0 *
(1-3)	7 6	400	6	600	100	0 *
(2-4)	7 6	400	5	600	100	1
(4-5)	5 4	1000	4	1350	87.5	0 *
(3-5)	10 9	500	5	800	50	4
(3-6)	7 6	400	6	500	100	0 *
(5-7)	5	1000	3	1400	200	2
(6-7)	8	500	5	950	150	3



Crashing Limit (for act. with min slope) = 3, 2

Compression Limit = {3, 2} = 2.

The critical Paths are:

1 → 2 → 4 → 5 → 7 *only possible* (5, 7)

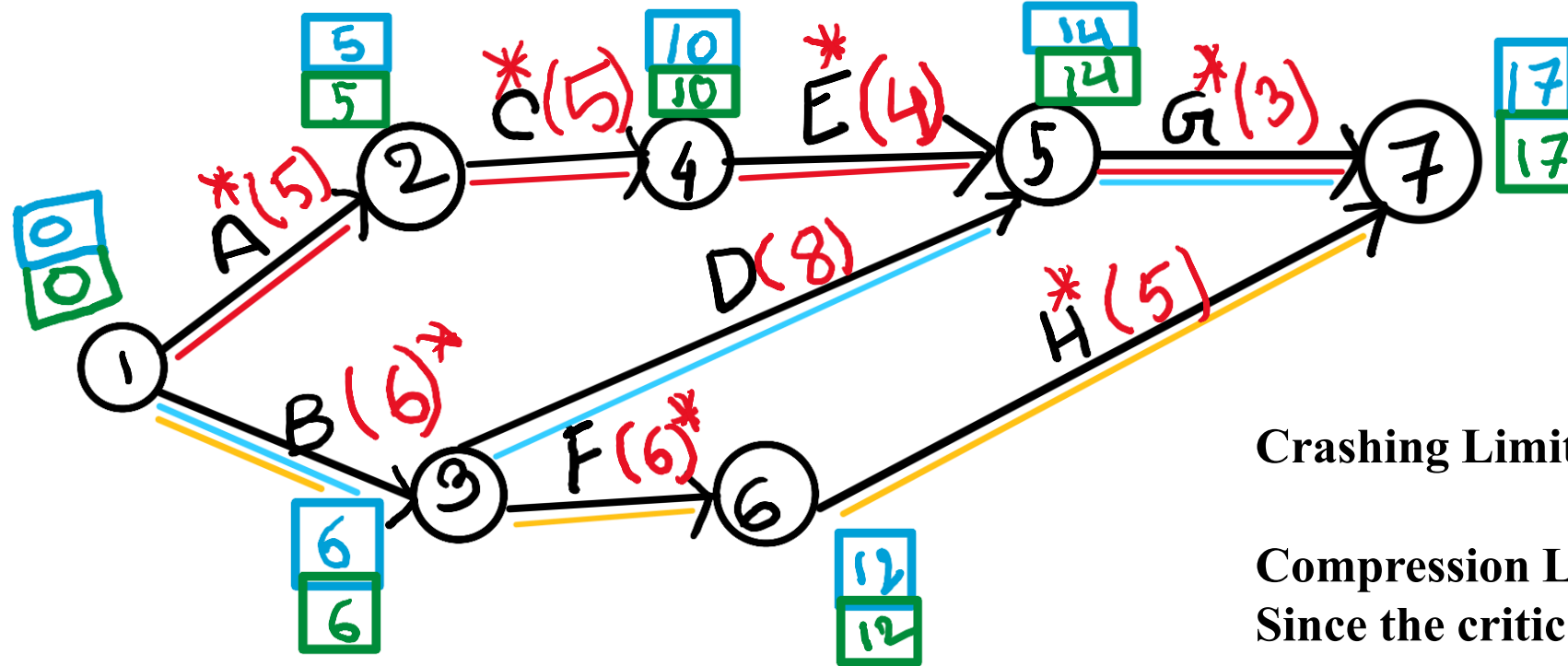
1 → 3 → 5 → 7 *choose activity* (3, 5)

1 → 3 → 6 → 7 *only possible* (6, 7)

Slope of (5, 7) < slope of (5, 7) + slope of (3, 5)

Therefore, we can reduce the duration of activity G(5, 7) and H(6, 7) by 2 days.

Activity	T_N	C_N	T_C	C_C	Slope	Crash Limit
(1-2)	6 5	300	5	400	100	0 *
(1-3)	7 6	400	6	600	100	0 *
(2-4)	6 5	400	5	600	100	0 *
(4-5)	5 4	1000	4	1350	87.5	0 *
(3-5)	9 8	500	5	800	50	3
(3-6)	7 6	400	6	500	100	0 *
(5-7)	5	1000	3	1400	200	2
(6-7)	8 7	500	5	950	150	2



Project Completion Time 17 days

Project Cost = 5750 + 200 + 150
= 6100

Crashing Limit (for act. with min slope) = 3, 2

Compression Limit = {3, 2} = 2.

Since the critical Paths

1 → 2 → 4 → 5 → 7

1 → 3 → 6 → 7

And

Have reached to their crashed duration.

Therefore, it is optimal crashing (atleast one critical path reached to its crashed time)

Optimal Project Completion Time 17 days

Optimal Project Cost = 6100

Activity	T_N	C_N	T_C	C_C	Slope	Crash Limit
(1-2)	6 5	300	5	400	100	0 *
(1-3)	7 6	400	6	600	100	0 *
(2-4)	6 5	400	5	600	100	0 *
(4-5)	5 4	1000	4	1350	87.5	0 *
(3-5)	9 8	500	5	800	50	3
(3-6)	7 6	400	6	500	100	0 *
(5-7)	5 3	1000	3	1400	200	0 *
(6-7)	7 5	500	5	950	150	0 *